

RAJARSHI MAITI

+49 155 1045799 • Bonn, Germany • rmaiti7@gmail.com • [rmaiti01.github.io](https://github.com/rmaiti01) • github.com/rmaiti01

PROFILE

Graduate mathematician (Number Theory, University of Bonn, current GPA 1.4/1.0) building infrastructure for AI-assisted formal mathematics. **Most recent public artifact: a CI-verified Lean 4 formalization of Erdős Problem #1026 (~330 readable lines vs. an AI prover's 3,658), with a gap-by-gap referee report of both informal proofs and a root-caused atlas of automation failures** (github.com/rmaiti01/weighted-erdos-szekeres). At ProofTree I work on content-addressable proof-state identity, syntactic canonicalisation, and semantic-faithfulness checking for AI-generated Lean 4, and I direct a Lean training programme for graduate mathematicians. Graduate coursework in dependent type theory, HoTT, and the Curry–Howard correspondence with Prof. Floris van Doorn (lead Mathlib contributor); research background in analytic number theory at the University of Oxford under Dr. Thomas Bloom (Royal Society Research Fellow). I work natively with AI coding agents and verify their output against the mathematics and the kernel. References available from Prof. Yang-Hui He (London Institute for Mathematical Sciences) and Dr. Thomas Bloom (Manchester/Oxford).

SELECTED FORMALIZATION WORK

Weighted Erdős–Szekeres: Erdős Problem #1026, Formalized and Refereed 2026
Independent Lean 4 development • github.com/rmaiti01/weighted-erdos-szekeres • zero sorries, CI-enforced axiom audit

- **Informal → formal translation at human scale.** Formalized the rectangle-packing resolution of Erdős #1026 (resolved Dec. 2025 by the AI prover Aristotle, then by a human in four sentences an hour later) in ~330 lines of readable, structured Lean 4 (Mathlib v4.29.0) — the theorem Aristotle’s machine proof establishes in 3,658 generated lines. Reproducible: pinned toolchain, CI on every push, kernel-checked axiom audit (`#print axioms` reports only the three standard axioms).
- **Gap analysis of informal proofs.** Refereed both proofs sentence-by-sentence (INFORMAL.md): seven hidden lemmas and unstated assumptions excavated from the four-sentence human proof — including an entire missing Mathlib theorem concealed behind the single word “with” — and six from the AI-found proof, including silent uses of positivity and distinctness.
- **Formalizability assessment as a deliverable.** Priced both proofs’ formalization costs step-by-step and selected the ~3–4× cheaper one — demonstrating that in each “morally the same” proof, the step the informal text spends the fewest words on is the most expensive to formalize.
- **Proof architecture and lemma design.** Decoupled the order-carrying map from the weight map, making dualization free and the main theorem strictly stronger than the source problem: classical Erdős–Szekeres recovered as the unit-weight corollary. Definitions ship their own API lemmas so no caller faces higher-order unification.
- **Automation failure analysis.** Failure atlas root-causing every place tactics or the library fell short (simp normal-form mismatches, higher-order unification failures, missing lemmas), with hard subgoals extracted to a standalone benchmark set for provers; surfaced three concrete, PR-able Mathlib gaps.

FORMAL VERIFICATION & PROOF-STATE INFRASTRUCTURE

ProofTree — Proof-State Infrastructure for AI Theorem Proving 2026 – Present
Co-Founder • Y Combinator Startup School 2026 (India), \$25k AI credits • Microsoft for Startups

- **Proof-state identity layer.** Designed a content-addressable system that separates three independent dimensions of a Lean proof state (tier: kind of claim; verified: whether checked; source: provenance) instead of collapsing them into one hash.
- **Syntactic vs. definitional equivalence.** Built a frozen syntactic canonicaliser (α -renaming, hypothesis and multi-goal sorting, `v1:<sha256>` addressing). Verified equivalence is layered as an explicit edge between states rather than a silent hash-merge.
- **Drift checker (in progress).** Evaluates whether AI-generated Lean is semantically faithful to its informal source rather than merely compilable, targeting formalisation drift in autoformalisation pipelines.
- **LeanDojo corpus instrumentation.** Tooling that probes the raw surface form of proof states in the LeanDojo Benchmark 4 corpus: exact codepoints, notation, source metadata.
- **Stack:** Lean verification backend, Next.js, Supabase; AI agents (Claude Code, Cursor) used natively throughout. App-layer code open source; collected process-trace dataset proprietary.

Director, Lean Formalisation Programme (ProofTree) 2026 (In Progress)
Bonn / Remote

- Designing and running a 10-week intensive programme taking graduate mathematicians with zero Lean experience to library-quality Lean 4 contributions and completed Mathlib PR cycles.
- Curriculum developed with input from Michael Rothgang (Mathlib contributor, University of Bonn), Mario Carneiro (Lean core contributor and Mathlib architect), and Yaël Dillies (co-organiser, Mathlib Reviewer Bootcamp 2026).
- Shortlisted for Renaissance Philanthropy’s AI for Math grant (backed by XTX Markets); major academic publishing house confirmed as co-sponsor.

TECHNICAL SKILLS

- **Formal Verification:** Lean 4: full informal-to-formal developments (Erdős #1026), proof-state systems, canonicalisation, LeanDojo tooling; graduate coursework in dependent type theory, HoTT, Curry–Howard; active in the Lean/Mathlib community.
- **Proof Engineering:** refereeing informal proofs for gaps and hidden assumptions; formalizability cost assessment; proof decomposition and API/lemma design; statement-faithfulness auditing; root-cause analysis of automation failures;

- benchmark extraction.
- **AI Engineering:** native development with agentic coding tools (Claude Code, Cursor); structuring problems so agents succeed; evaluating and correcting agent output on mathematical and architectural decisions.
 - **Programming:** Python (data processing, benchmarking pipelines); TypeScript / Next.js; Supabase.
 - **Mathematical Evaluation:** proof annotation, fine-grained correctness decomposition, STEM OCR review, benchmark curation for AI mathematical reasoning (professional: Mercor, Alignerr).
 - **Typesetting:** $\text{\LaTeX}$ (advanced): papers, grant proposals, structured proof documents.

EDUCATION

MSc Mathematics, University of Bonn 2025 – 2027 (Expected)
Specialisation: Number Theory • Current GPA 1.4/1.0 • Bonn, Germany

- Logic of Proof Assistants with Prof. Floris van Doorn (lead Mathlib contributor): λ -calculus, dependent type theory, HoTT, Curry–Howard correspondence.
- Further graduate coursework: Advanced Analysis, Abstract Algebra, Topology, Probability Theory.
- Presented graduate research seminar: “Smooth Circle Method by Heath-Brown.”

BSc Mathematics, Queen’s University Belfast 2019 – 2022
Pure Mathematics • Top 10% of cohort (Year 2) • Belfast, UK

- Abstract Algebra, Real & Complex Analysis, Topology, Algebraic Topology, Number Theory, Discrete Mathematics, Probability & Measure Theory (graduate level).

RESEARCH EXPERIENCE

Summer Research Student, University of Oxford Jun – Aug 2022
Supervisor: Dr. Thomas Bloom (Royal Society Research Fellow) • Oxford, UK

- Investigated gaps between sums of two squares using analytic number theory and probabilistic methods; rigorous proofs requiring careful lemma decomposition and structured argumentation, translating intuitive reasoning into explicit, verifiable steps.

ML for Analytic Number Theory, with Prof. Yang-Hui He (LIMS) 2026 – Present
Early-stage collaboration

- Exploring whether optimisation-type problems arising in the circle method and gaps between sums of two squares are amenable to machine-learning methods; scoping a write-up from current understanding.

Summer Research Student, Independent Research Jun – Aug 2021
Supervisor: Dr. Tiago Jardim da Fonseca (Postdoc, University of Oxford) • Remote

- Explored special values of zeta and L -functions; identified and articulated hidden assumptions and implicit reasoning steps in mathematical arguments.

PROFESSIONAL EXPERIENCE

Math AI Trainer (multiple roles), Alignerr & Mercor Mar 2026 – Present
Evaluation of mathematical reasoning for leading AI labs • Remote

- Benchmark undergraduate- and graduate-level mathematics for frontier AI training; decompose arguments into evaluable steps with fine-grained annotations on correctness, logical structure, and difficulty; structured STEM OCR review; multi-turn AI system comparison.

Graduate Student Assistant, Mathematical Institute, University of Bonn 2025 – Present
Bonn, Germany

Quantitative Trading Camp (Spring Programme), Jane Street Apr 2024
Fully sponsored, highly competitive programme • London, UK

- Intensive training in probability, logical reasoning, and decision-making under time constraints; invited to the Quantitative Trading Internship interview process based on performance.

AWARDS & SELECTED INVITATIONS

- **Lean–LMFDB Bridge Workshop** (University of East Anglia, Norwich): fully funded invitation; L -functions and formalisation. Hosts: Dr. Christopher Birkbeck, David Ang. Jun – Jul 2026
- **AI × Mathematics Conference** (ICMS Edinburgh): selected for full sponsorship. Feb 2026
- **Microsoft for Startups:** ProofTree accepted into the programme. 2026
- **Y Combinator AI Startup School:** selected participant; \$25,000 in startup credits. 2025
- **DAAD STIBET Scholarship** (German Federal Foreign Office). Oct 2025

LEADERSHIP & OUTREACH

Director, BMUCO, Global Science Non-Profit (bmuco.org) 2019 – Present
Pivoting to a registered e.V. (Germany)

- Co-organised the Abel Symposium 2025 with the Norwegian Academy of Science and Letters and the Hausdorff Center for Mathematics, Bonn; host public interviews with Fields Medalists and Nobel Laureates — content reaching 20,000+ students globally.